
Dual Certificates for Agent Audit: Separating Structural Unrecoverability from Decision Relevance

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Abstract

1 Auditing a deployed language-model agent requires two separable quantities: how
2 much operative state escapes the recorded trace, and how much of that residual state
3 drives behavior. We introduce a dual-certificate protocol. The static certificate $\varepsilon_{\text{state}}^{\text{UB}}$
4 upper-bounds residual hidden-state entropy by a min-cut on untraced channels.
5 The dynamic certificate $\delta_{\text{act}}^{\text{LB}}$ lower-bounds residual decision relevance through
6 an admissible probe taxonomy—replay, intervention, proxy—under conditional
7 data processing. The two axes are independent. In ReAct experiments, logging
8 ablates the static bound from 16,464 to 0 bits; controlled replay separates dormant
9 calculator from active planning tasks under the same topology as a soft policy shift
10 (0.0163 bits, 95% CI [0.0124, 0.0208]) with argmax tool selections unchanged.
11 Indexing $\delta_{\text{act}}^{\text{LB}}$ over hidden-channel coordinates produces an activation profile. On
12 an LLaDA denoising trajectory, perturbations stay near the floor through early
13 steps and rise at the final binding step (0.110 bits, 95% CI [0.052, 0.234]). On
14 a multi-agent communication edge, swapping a worker’s private report gives
15 0.901 bits, 95% CI [0.873, 0.928]. A Lean 4 artifact mechanizes the certificate
16 reductions (static cut-sum bound, conditional-DPI dynamic certificate), the linear
17 chain case study, the autoregressive zero-cut impossibility, and an identifiability-gap
18 construction from Mathlib first principles, with only the cut-set capacity inequality
19 remaining as an external structural premise.

20 1 Introduction

21 Consider a ReAct-style agent shipped with an unlogged scratchpad of capacity 16,384 bits per turn.
22 A regression test on output traces shows no behavioral change, yet the same trace under controlled
23 replay with the scratchpad disabled reveals a 0.0163-bit shift in the tool-token probability distribution
24 on planning tasks. The disagreement is not a probing artifact. It is a structural feature of the audit:
25 the recorded trace fails to distinguish two qualitatively different systems, one whose hidden capacity
26 is dormant on the test distribution and one whose hidden capacity is active. Identifying which holds
27 is the audit task.

28 How much internal state escapes the trace, and how much of that drives the next action, are distinct
29 quantities. Treating them as a single “hiddenness” judgment produces something neither computable
30 nor diagnostic: a system with a large logging gap but dormant hidden drive is indistinguishable from
31 one with a small gap but strong drive, though the audit implications differ.

32 We propose a dual-certificate protocol that answers each question separately. The static certificate
33 $\varepsilon_{\text{state}}^{\text{UB}}$ upper-bounds residual entropy $H(S_t \mid \tilde{T}_t)$ from deployment topology alone, using a cut-set
34 bound on untraced channels. The dynamic certificate $\delta_{\text{act}}^{\text{LB}}$ lower-bounds residual decision relevance
35 $I(S_t; A_t \mid \tilde{T}_t)$ via one of three probe classes (replay, intervention, or proxy) depending on the
36 auditor’s access regime. The two certificates are independent axes: a system with large $\varepsilon_{\text{state}}^{\text{UB}}$ and

37 small $\delta_{\text{act}}^{\text{LB}}$ has unrecoverable but dormant state; a system with both large has unrecoverable and active
 38 state. Both are auditable; neither is captured by a single scalar.

39 The dynamic axis can also be indexed. Instead of asking only whether residual state matters, an
 40 auditor can ask *where* it matters: which module, denoising step, or communication edge carries
 41 decision relevance. This turns the certificate from a scalar diagnostic into an activation profile over
 42 the hidden-state geometry of the deployment.

43 Conditioning on $\tilde{T}_t$ is part of the protection, not a nuisance term. If a model writes a steganographic
 44 or compressed rationale into the recorded trace, that information is in $\tilde{T}_t$ and is conditioned away
 45 by $I(S_t; A_t \mid \tilde{T}_t)$. The dynamic certificate measures residual decision relevance beyond the logged
 46 trace; it does not count hiddenness that has already been made visible to the audit record.

47 **Contributions.** We define a dual-certificate framework $(\varepsilon_{\text{state}}^{\text{UB}}, \delta_{\text{act}}^{\text{LB}})$ for agent audit. Proposition 1
 48 gives a topological upper bound on residual hidden-state entropy via a min-cut on untraced channels.
 49 Proposition 2 is not a new information-theoretic inequality; it is the audit reduction that turns
 50 conditional DPI into three admissible lower-bound probes—replay, intervention, and proxy—for
 51 residual decision relevance.

52 We evaluate the protocol on three residual-state geometries: a ReAct scratchpad module, a diffusion-
 53 LM denoising trajectory, and a multi-agent private-report edge. The same algebra applies across
 54 settings; only the index changes (module, denoising step, communication edge). Appendix cal-
 55 ibrations cover read-only proxy estimation and a synthetic ground-truth setting where the static
 56 bound matches true hidden-state entropy. A Lean 4 artifact mechanizes Corollary 2, Proposition 1,
 57 Proposition 2, the linear chain case study, and the Identifiability Gap theorem from Mathlib first
 58 principles, with only the cut-set capacity bound remaining as an external structural premise (see §D.1
 59 for the precise boundary).

60 2 Related Work

61 **Probing, patching, and causal abstraction.** Causal abstraction [Geiger et al., 2021], latent-
 62 knowledge elicitation [Burns et al., 2022], nullspace projection [Ravfogel et al., 2020], and amnesic
 63 probing [Elazar et al., 2021] are natural sources of proxy-style evidence: when they expose a probe
 64 variable $Z_t = \varphi(S_t)$ and the required conditional assumptions hold, its MI with action can be
 65 interpreted through the proxy certificate. Causal tracing/ROME [Meng et al., 2022], representation
 66 engineering [Zou et al., 2023], and activation addition [Turner et al., 2023] provide intervention-style
 67 evidence when the patch affects action only through S_t .

68 **Black-box audit and network information theory.** Property testing [Goldreich et al., 1998] and
 69 black-box safety auditing [Casper et al., 2024, Su et al., 2024] study what can be inferred from
 70 outputs alone. The distinction here is that output-only access can support behavioral tests, but it does
 71 not by itself provide a lower bound on δ_{act} ; $\varepsilon_{\text{state}}^{\text{UB}}$ remains computable from topology when structural
 72 access is available. The static certificate proof applies a cut-set upper bound [El Gamal and Kim,
 73 2011] to the time-unrolled DAG (full derivation in Appendix A).

74 **Diffusion language-model agents.** LLaDA is a large-scale masked diffusion language model with
 75 bidirectional denoising and instruction-following [Nie et al., 2025]. Recent agent work further studies
 76 diffusion LMs in multi-step decision and tool-use workflows, including matched comparisons to
 77 autoregressive agents [Zhen et al., 2026]. Intermediate denoising latents in these systems are a natural
 78 hidden channel for the dynamic certificate.

79 3 Setup and Audit Regime

80 Standard information-theoretic notation follows Cover and Thomas [2006]. All logarithms are base
 81 2; entropy and mutual information are reported in bits.

82 At step t : *full trace* T_t (all intermediate activations), *recorded trace* $\tilde{T}_t \subseteq T_t$ (auditor-visible),
 83 *operative state* S_t (internal computation), *action* A_t (next token), *unrecorded sources* U_t (exogenous
 84 inputs not in $\tilde{T}_t$).

85 3.1 Two Core Audit Quantities

86 **Definition 1** (Core Audit Quantities). *The two quantities targeted by the dual certificate are:*

- 87 • **Structural unrecoverability:** $\varepsilon_{state} := H(S_t | \tilde{T}_t)$, the residual entropy of the operative state
88 given the visible trace.
- 89 • **Residual decision relevance:** $\delta_{act} := I(S_t; A_t | \tilde{T}_t)$, the mutual information between
90 operative state and next action given the visible trace.

91 *Neither quantity is directly computable in deployment: ε_{state} requires recovering the full internal-state
92 distribution, and δ_{act} requires observing S_t . The dual-certificate framework targets:*

- 93 • a structural upper bound $\varepsilon_{state}^{UB} \geq \varepsilon_{state}$, computed from deployment topology (§4);
- 94 • an empirical lower bound $\delta_{act}^{LB} \leq \delta_{act}$, estimated via probe variables (§5).

95 *The reported audit pair is $(\varepsilon_{state}^{UB}, \delta_{act}^{LB})$.*

96 *Neither quantity is identifiable from behavioral observations alone: different causal graphs can
97 produce identical joint distributions over $(\tilde{T}_t, A_t)$ pairs, so observing outputs does not fix $H(S_t | \tilde{T}_t)$
98 or $I(S_t; A_t | \tilde{T}_t)$. The static certificate addresses the first with structural access to topology; the
99 dynamic certificate addresses the second with gray-box probe access.*

100 **Remark 1** (Complementarity and Inheritance). *By the data processing inequality, $\delta_{act} \leq \varepsilon_{state}$. The
101 certificates inherit this ordering at their respective bound directions but remain independent axes of
102 audit evidence:*

- 103 • If $\varepsilon_{state}^{UB} = 0$ (fully logged architecture), then $\varepsilon_{state} = 0$ and $\delta_{act} = 0$; any valid lower bound
104 satisfies $\delta_{act}^{LB} = 0$.
- 105 • If $\delta_{act}^{LB} = 0$ under the best available probe, hidden state may still exist: ε_{state}^{UB} can be large
106 while the current task does not activate the hidden capacity (dormant unrecoverable state).
107 The contrapositive ($\delta_{act}^{LB} > 0 \Rightarrow \varepsilon_{state}^{UB} > 0$) follows from the DPI ordering.

108 *The audit interpretation is therefore two-dimensional: deployment teams must report both the
109 residual channel budget and the strongest admissible behavioral probe, not collapse them into a
110 single hiddenness score.*

111 3.2 Audit Access Modes

- 112 1. *Structural access.* The auditor has the deployment architecture, logging manifest, and
113 protocol budgets. This is sufficient to compute ε_{state}^{UB} from topology (§4) but gives no
114 dynamic lower bound.
- 115 2. *Controlled replay access.* The auditor can rerun the same system under wild and replayed
116 states while holding the visible trace fixed. This supports replay certificates when the replay
117 mechanism affects the action only through missing state recovery.
- 118 3. *Proxy or intervention access.* The auditor can read a proxy $Z_t = f(S_t)$ or perturb a hidden
119 module. These support conditional-DPI lower bounds on δ_{act} (§5).

120 All three modes are instantiated in the controlled ReAct experiments below. Paraphrase- or format-
121 based black-box replay is not used as evidence for internal decision relevance.

122 4 Static Certificate: Structural Upper Bound via Untraced-Channel Capacity

123 The static certificate upper-bounds $\varepsilon_{state} = H(S_t | \tilde{T}_t)$ from deployment topology. Information not
124 in the visible trace that can influence S_t must travel through unlogged channels. Bounding those
125 channels' worst-case capacity bounds the hidden-state entropy.

4.1 Time-Unrolled Deployment Graph and Component Budgets

Let $G_t = (V_t, E_t)$ be the time-unrolled DAG of the deployment up to step t . Nodes are state updates, tool calls, memory reads/writes, messages; edges are information channels. Three object classes: unrecorded sources U_t (retrieval results, external memory, unlogged messages), the visible trace $\tilde{T}_t$, and the operative state S_t . Let $\mathcal{C}_{\text{unlogged}}$ be all reachable edge cuts separating U_t from S_t .

A set of edges is *software-orthogonal* if, conditioned on $\tilde{T}_t$, the channel outputs factor: $p(\{y_e\}_{e \in E'} \mid \{x_e\}, \tilde{T}_t) = \prod_{e \in E'} p(y_e \mid x_e, \tilde{T}_t)$. This holds when unlogged channels use independent API calls or separate memory regions (see Appendix A for non-orthogonal remediation).

Lemma 1 (Additive decomposition). *Under software orthogonality with per-edge budgets c_e , the induced cut capacity decomposes: $C_{\text{cut}}(\Omega) \leq \sum_{e \in E(\Omega, \Omega^c)} c_e$.*

When software orthogonality fails (for example, when two unlogged channels share a hidden state variable) the per-edge sum $\sum_{e \in E(\Omega, \Omega^c)} c_e$ still upper-bounds $C_{\text{cut}}(\Omega)$ by sub-additivity of mutual information across coupled channels: $I(X_\Omega; Y_{\Omega^c} \mid \tilde{T}_t, X_{\Omega^c}) \leq \sum_e I(X_e; Y_e \mid \tilde{T}_t) \leq \sum_e c_e$. Orthogonality is required for *tightness* of the additive decomposition, not for validity of the upper bound (formal treatment in Appendix A).

Per-edge budgets c_e are read from interface specifications: $K \log |\mathcal{V}|$ for a K -token text channel over vocabulary $\mathcal{V}$, $d \log Q$ for a d -dimensional quantized state, $\log |\Omega_{\text{states}}|$ for a discrete scheduler. The discretization parameter Q is an auditor-side choice (we use $Q = 256$, 8-bit audit discretization throughout); tighter or coarser discretizations shift the bound proportionally without changing the structural claim. All per-edge bounds are worst-case (maximum-entropy) bounds; tighter empirical bounds only improve the certificate.

Proposition 1 below has three terms. $\varepsilon_{\text{nominal}} = H(S_t \mid T_t)$ is the residual uncertainty after a complete internal trace; for a fully instrumented software agent this is zero. The cut term budgets information that can still enter S_t through unlogged interfaces. The directed-MI notation prices that worst-case flow formally; the implemented certificate needs only per-edge budgets and a min-cut on the unlogged subgraph.

Proposition 1 (Static Certificate via Cut-Set Bound). *For any cut Ω separating U_t from S_t with induced capacity $C_{\text{cut}}(\Omega) = \sup_{p(X_\Omega \mid \tilde{T}_t)} I(X_\Omega \rightarrow Y_{\Omega^c} \mid \tilde{T}_t, X_{\Omega^c})$,*

$$H(S_t \mid \tilde{T}_t) \leq \underbrace{H(S_t \mid T_t)}_{\varepsilon_{\text{nominal}}} + \min_{\Omega} C_{\text{cut}}(\Omega).$$

Setting $\varepsilon_{\text{state}}^{\text{UB}} := \varepsilon_{\text{nominal}} + \min_{\Omega} C_{\text{cut}}(\Omega)$ gives $\varepsilon_{\text{state}} \leq \varepsilon_{\text{state}}^{\text{UB}}$.

Here X_Ω denotes the input signals entering edges crossing the cut, Y_{Ω^c} the corresponding output signals, and X_{Ω^c} the boundary state on the sink side (formal definitions in Appendix A).

Corollary 1 (Additive form). *Under software orthogonality, $\varepsilon_{\text{state}}^{\text{UB}} = \varepsilon_{\text{nominal}} + \min_{C \in \mathcal{C}_{\text{unlogged}}} \sum_{e \in C} c_e$, a discrete min-cut on the unlogged subgraph.*

Proof sketch: Chain-rule gives $H(S_t \mid \tilde{T}_t) = H(S_t \mid T_t) + I(S_t; M_t \mid \tilde{T}_t)$ where $M_t = T_t \setminus \tilde{T}_t$. Every path from U_t to S_t crosses a cut Ω , so M_t is d -separated from S_t by the cross-cut signals; DPI yields $I(S_t; M_t \mid \tilde{T}_t) \leq C_{\text{cut}}(\Omega)$. Minimizing over cuts and substituting Lemma 1 completes the proof (full derivation in Appendix A).

Corollary 2 (Autoregressive zero-cut). *If the system is a strict autoregressive core and $\tilde{T}_t$ includes the full context window, then $\mathcal{C}_{\text{unlogged}} = \emptyset$, every cut has zero induced capacity, and $\varepsilon_{\text{state}}^{\text{UB}} = 0$.*

5 Dynamic Certificate: Decision Relevance via Conditional DPI

The dynamic certificate targets $\delta_{\text{act}} := I(S_t; A_t \mid \tilde{T}_t)$. Since S_t is unobservable, we specify probes whose measured variables satisfy $X_t \rightarrow S_t \rightarrow A_t$ given $\tilde{T}_t$. Conditional DPI supplies the correctness argument: $I(X_t; A_t \mid \tilde{T}_t) \leq \delta_{\text{act}}$.

Proposition 2 (Probe Certificates from Conditional DPI). *For any admissible probe variable X_t with $X_t \rightarrow S_t \rightarrow A_t$ given $\tilde{T}_t$, the probe measurement $I(X_t; A_t \mid \tilde{T}_t)$ is a valid lower-bound certificate for δ_{act} .*

Replay certificates. Let $R_t \in \{\text{wild}, \text{replay}\}$ indicate whether a missing state fragment is reconstructed under the same visible trace $\tilde{T}_t$ (controlled re-execution, not paraphrase variation). Then $I(R_t; A_t \mid \tilde{T}_t) = \text{JS}(P_{\text{wild}}, P_{\text{replay}} \mid \tilde{T}_t) \leq \delta_{\text{act}}$. Discrete action spaces admit direct empirical JS estimation over either realized actions or model-reported action probabilities; the experiments below state which is used.

Intervention certificates. Let ξ_{hidden} be an exogenous perturbation to a suspected unlogged module (cache, memory buffer, scratchpad). If $\xi_{\text{hidden}} \rightarrow S_t \rightarrow A_t$ given $\tilde{T}_t$, then $I(\xi_{\text{hidden}}; A_t \mid \tilde{T}_t) \leq \delta_{\text{act}}$. Discrete action spaces admit direct conditional action-distribution shift estimates, such as JS divergence; for continuous action spaces, InfoNCE [van den Oord et al., 2018] or MINE [Belghazi et al., 2018] can be used. Requires gray-box perturbation access; the most direct causal probe of the three.

Proxy certificates. Let $Z_t = f(S_t)$ be a readable coarsening of the operative state (tool-logit projection, attention summary). Since $Z_t \leftarrow S_t \rightarrow A_t$ given $\tilde{T}_t$, conditional DPI yields $I(Z_t; A_t \mid \tilde{T}_t) \leq \delta_{\text{act}}$. We estimate this quantity by fitting predictors $p(A_t \mid \tilde{T}_t)$ and $p(A_t \mid \tilde{T}_t, Z_t)$ and taking their cross-entropy gap, $\hat{I}(Z_t; A_t \mid \tilde{T}_t)$.

Aggregation. A single certificate class can miss some causal paths. Define $\delta_{\text{act}}^{\text{LB}} := \max\{I(R_t; A_t \mid \tilde{T}_t), I(\xi_{\text{hidden}}; A_t \mid \tilde{T}_t), I(Z_t; A_t \mid \tilde{T}_t)\}$. By monotonicity of the maximum, the aggregate remains a valid lower bound on δ_{act} .

Activation profiles. The scalar certificate is the maximum over an indexed family of admissible probes. Keeping the index gives an *activation profile*

$$\Delta(j) := I(X_t^{(j)}; A_t \mid \tilde{T}_t),$$

where j may denote a scratchpad field, a denoising step, an activation layer, or a communication edge. The static certificate identifies where residual capacity can reside; the activation profile identifies where that capacity is behaviorally used. The experiments below keep this algebra fixed while changing the index: module in ReAct, time in the diffusion LM, and communication edge in the multi-agent setting.

6 Empirical Diagnostics

The static ablation (§6.1) computes the residual bit budget from topology. ReAct intervention and controlled replay (§6.2) show the same unlogged scratchpad dormant on calculator tasks and active on planning tasks. The LLaDA and multi-agent probes index $\Delta(j)$ by denoising step and peer-report edge, respectively (§6.3–6.4). Read-only proxy and synthetic ground-truth calibrations are in Appendices C–E. All pipelines run on a single M4 Max workstation.

6.1 Static Certificate: Min-Cut and Logging Ablation

We extract the time-unrolled DAG from Qwen2.5-7B-Instruct ReAct execution traces on tool-selection tasks (Fig. 1, left). Per-edge budgets c_e are assigned from interface specifications (§4); the min-cut on unlogged edges gives $\varepsilon_{\text{state}}^{\text{UB}}$. Varying logging levels recomputes the min-cut; Table 1 reports the stepwise reduction from 16,464 to 0 bits. The dominant bottleneck is the scratchpad-read path (16,384 bits). The same static computation also supplies the coordinate system for the diffusion-LM and multi-agent dynamic checks that follow.

These cuts put the later dynamic probes on a common map: a scratchpad module in ReAct, a denoising interface in LLaDA, and a peer-state edge in the multi-agent system.

6.2 Dynamic Causal Certificates: Intervention and Controlled Replay

We separate the two axes with a dormant/active task split under one topology. The unlogged scratchpad is present in both cases: irrelevant for calculator tasks, necessary for planning tasks. Intervention perturbs the scratchpad (masking or replacement); controlled replay removes it entirely

Logging level	$\varepsilon_{\text{state}}^{\text{UB}}$ (bits)	Bottleneck
ReAct, output only	16,464	scratchpad_read (16384) + router_logits (80)
ReAct, +log router	16,384	scratchpad_read (16384)
ReAct, +log scratchpad	0	none
ReAct, full instrumentation	0	none
LLaDA-8B-Instruct ($K=128$)	32,768	denoising interface
Multi-agent, all logged	0	none
Multi-agent, peer report unlogged	32,768	peer_report $\rightarrow S_t$

Table 1: Static certificate geometry. Adding ReAct logs monotonically removes residual cut capacity; the same min-cut analysis localizes capacity in a scratchpad module, denoising interface, or peer-report edge. The LLaDA bound is independent of K because repeated reuse of the same bounded latent interface does not increase the one-step information bottleneck.

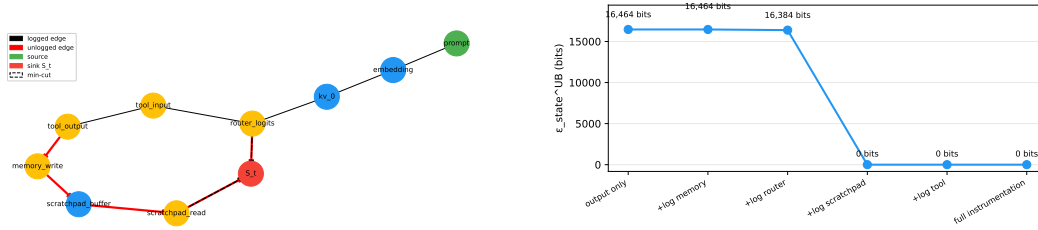


Figure 1: Left: Time-unrolled DAG extracted from Qwen2.5-7B-Instruct ReAct execution (logical edges omitted for clarity). Red edges are unlogged. Dashed edges form the min-cut. Right: Logging ablation curve. Adding instrumentation at the bottleneck drops $\varepsilon_{\text{state}}^{\text{UB}}$ stepwise to zero.

216 while holding $\tilde{T}_t$ fixed. Table 2 reports JS divergence over realized tool actions. Table 3 reports JS
217 divergence over the model’s soft tool-token probability distribution; for the headline replay contrast,
218 the argmax tool counts are unchanged, so the evidence is a policy-probability shift rather than a
219 realized-tool flip. Both certificates are zero on dormant tasks and positive on active tasks. The
220 topology, and therefore $\varepsilon_{\text{state}}^{\text{UB}}$, is unchanged.

221 The pair of tables isolates the transition from high- ε /low- δ to high- ε /high- δ : the hidden channel is
222 structurally present in both task splits, but it becomes action-relevant only on planning tasks.

223 Under this fixed topology, the unlogged scratchpad contributes 16,384 bits to $\varepsilon_{\text{state}}^{\text{UB}}$ in both task splits,
224 matching the min-cut from the extracted ReAct DAG. Intervention and replay are zero on calculator
225 tasks and positive on planning tasks. Structural unrecoverability is set by the deployment boundary
226 and logging policy; residual decision relevance depends on which hidden channels the task uses.

227 6.3 Temporal Certificate Analysis on a Diffusion LM

228 ReAct gives a module-level contrast: the scratchpad is either used or not used. A diffusion LM gives
229 a temporal contrast. The hidden channel exists throughout the K -step denoising trajectory, but its
230 decision relevance need not be constant across steps.

231 We use LLaDA-8B-Instruct with $K=10$ denoising steps on tool-selection prompts. A Gaussian
232 perturbation ($\sigma=5.0$) is applied to layer-1 activations at steps $\{2, 4, 6, 8, 10\}$; a late-layer perturbation
233 is tracked as a broad sensitivity control. Each point uses $n=20$ trajectories and a trajectory-block
234 bootstrap with 1,000 resamples.

235 The pattern is late-binding. Steps 2–8 remain near the floor for the target layer (0.001–0.006 bits),
236 while step 10 rises to 0.110 bits with 95% CI $[0.052, 0.234]$. The control layer also rises at step 10
237 (0.092 bits, CI $[0.043, 0.186]$), so the sensitivity is global rather than confined to the chosen layer.
238 LLaDA carries hidden capacity throughout denoising; it becomes action-relevant when the denoising
239 state binds to a discrete tool choice.

Task	Perturbation	JS (bits; nats)	95% CI (bits)
Calculator (dormant)	mask 0.7	0.0000; 0.0000	[0, 0]
Planning (active)	mask 0.7	0.0215; 0.0149	[0.0094, 0.0410]
Calculator (dormant)	replace 1.0	0.0000; 0.0000	[0, 0]
Planning (active)	replace 1.0	0.0055; 0.0038	[0.0016, 0.0131]

Table 2: Dynamic intervention certificate under the same unlogged-scratchpad topology. Dormant calculator tasks show no detectable scratchpad influence; active planning tasks show positive action-distribution shift.

Task	Replay condition	JS (bits; nats)	95% CI (bits)
Calculator (dormant)	scratchpad removed	0.0000; 0.0000	[0, 0]
Planning (active)	scratchpad removed	0.0163; 0.0113	[0.0124, 0.0208]

Table 3: Controlled replay certificate over soft tool-token probability distributions. Wild episodes use the full hidden scratchpad; replay episodes use the same visible trace with the scratchpad removed. The dormant-to-active contrast mirrors the intervention result, although the headline replay contrast does not change argmax tool counts.

6.4 Multi-Agent Private-Report Intervention

The multi-agent setting changes the index again. Here the controller delegates evidence gathering to a worker, so the worker’s private report is not an auxiliary scratchpad but an epistemic edge in the decision graph. The worker sees a benchmark item and writes a short report; the controller sees only that report and chooses one API from `{calculator, search}`. The task pool combines 200 GSM8K arithmetic items [Cobbe et al., 2021] and 200 HotpotQA multi-hop fact questions [Yang et al., 2018] with seed 42.

The main intervention replaces the worker report with a report from the opposite task class while holding the controller protocol fixed. A neutral replay removes the report content, and a same-class shuffle replaces it with another report from the same task class. An oracle-tag condition directly names and flips the API label to measure the binary ceiling. For each task and condition, the controller is sampled $k=8$ times; labels are parsed as `{calculator, search, invalid}`; and $JS(P_{\text{base}}, P_{\text{condition}})$ is averaged with a task-block bootstrap.

The contrast is topological. In ReAct, the scratchpad matters only when the task needs it. In the multi-agent deployment, the controller’s decision is routed through the private worker report, so the active site is the communication edge itself. Same-class variation is local, neutral removal is substantial, and counterfactual evidence reaches 0.901 bits, close to the oracle ceiling.

7 Discussion

7.1 One Plot: Capacity, Then Activation

The three experiments follow a single logic. Static certificates map where hidden capacity can reside—scratchpad modules, denoising interfaces, peer-state edges. Dynamic certificates identify which of those sites affect actions. The same scratchpad is dormant on calculator tasks and active on planning tasks in ReAct; in LLaDA, action relevance concentrates at the final binding step; in the multi-agent deployment, delegation makes the private worker report the active edge.

Effect sizes index deployment topology. Under the same ReAct topology, calculator tasks leave the scratchpad dormant while planning tasks produce nonzero action-distribution shifts under intervention (0.0215 bits, CI [0.0094, 0.0410]) and controlled replay (0.0163 bits, CI [0.0124, 0.0208]). The LLaDA final binding step rises to 0.110 bits; the multi-agent counterfactual report edge rises to 0.901 bits. A hidden channel that supplements an already-determined visible trace has low $\delta_{\text{act}}^{\text{LB}}$; one that carries the decision has high $\delta_{\text{act}}^{\text{LB}}$. Under this protocol, a claim that an unobserved variable causally affects behavior must identify both a channel in $\varepsilon_{\text{state}}^{\text{UB}}$ and a probe in $\delta_{\text{act}}^{\text{LB}}$.

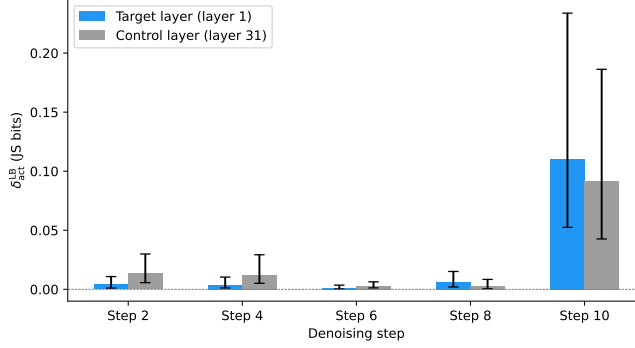


Figure 2: Temporal activation profile for LLaDA-8B-Instruct. Early denoising steps have low dynamic certificates; the final action-forming step has a much larger certificate for both target and control layers, indicating late-binding decision relevance rather than a module-local scratchpad effect.

Condition	$\delta^{\text{LB}}_{\text{act}}$ (JS bits)	95% CI (bits)
Neutral replay	0.358	[0.324, 0.392]
Counterfactual evidence	0.901	[0.873, 0.928]
Same-class shuffle	0.084	[0.060, 0.109]
Oracle tag upper-bound	1.000	[1.000, 1.000]

Table 4: Topological activation profile for the multi-agent deployment. Counterfactual evidence nearly saturates the binary action ceiling, while same-class shuffling remains much smaller. Hidden communication is therefore behaviorally active when the controller has delegated evidence gathering to the worker.

Acknowledgments

We thank the developers of Qwen2.5 and LLaDA for the open-weight models used in the ReAct and diffusion-LM experiments; the creators and maintainers of GSM8K and HotpotQA for benchmark data used in the multi-agent setting; and the Hugging Face Transformers, PyTorch, NumPy, scikit-learn, NetworkX, Matplotlib, Lean 4, Lake, and Mathlib communities for the software infrastructure supporting the experiments, figures, and formalization. Computations ran on a single Apple M4 Max workstation.

References

- Rudolf Ahlswede, Ning Cai, Shuo-Yen Robert Li, and Raymond W. Yeung. Network information flow. *IEEE Transactions on Information Theory*, 46(4):1204–1216, 2000. doi: 10.1109/18.850663.
- Mohamed Ishmael Belghazi, Aristide Baratin, Sai Rajeshwar, Sherjil Ozair, Yoshua Bengio, Aaron Courville, and Devon Hjelm. Mutual information neural estimation. In *Proceedings of the 35th International Conference on Machine Learning*, pages 531–540, 2018.
- Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. Discovering latent knowledge in language models without supervision. *arXiv preprint arXiv:2212.03827*, 2022. doi: 10.48550/arXiv.2212.03827.
- Stephen Casper, Carson Ezell, Charlotte Siegmann, Noam Kolt, Taylor L. Curtis, Benjamin Bucknall, Andreas Haupt, Kevin Wei, J  r  my Scheurer, Marius Hobbhahn, et al. Black-box access is insufficient for rigorous AI audits. In *The 2024 ACM Conference on Fairness, Accountability, and Transparency*, 2024. doi: 10.1145/3630106.3659037.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021. doi: 10.48550/arXiv.2110.14168.

295 Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, 2nd
296 edition, 2006.

297 Abbas El Gamal and Young-Han Kim. *Network Information Theory*. Cambridge University Press,
298 2011.

299 Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral
300 consequences of information removal from a representation. *Transactions of the Association for*
301 *Computational Linguistics*, 9:346–361, 2021.

302 Atticus Geiger, Hanson Lu, Thomas Icard, and Christopher Potts. Causal abstractions of neural
303 networks. In *Advances in Neural Information Processing Systems*, volume 34, pages 9574–9586,
304 2021.

305 Oded Goldreich, Shafi Goldwasser, and Dana Ron. Property testing and its connection to learning
306 and approximation. *Journal of the ACM*, 45(4):653–750, 1998.

307 Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual
308 associations in GPT. In *Advances in Neural Information Processing Systems*, 2022.

309 Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang, Jingyang Ou, Jun Hu, Jun Zhou, Yankai Lin, Ji-
310 Rong Wen, and Chongxuan Li. Large language diffusion models. *arXiv preprint arXiv:2502.09992*,
311 2025. doi: 10.48550/arXiv.2502.09992.

312 Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2nd edition,
313 2009.

314 Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. Null it out:
315 Guarding protected attributes by iterative nullspace projection. In *Proceedings of the 58th Annual*
316 *Meeting of the Association for Computational Linguistics*, pages 7237–7256, 2020.

317 Jingtong Su, Julia Kempe, and Karen Ullrich. Mission impossible: A statistical perspective on
318 jailbreaking LLMs. In *Advances in Neural Information Processing Systems*, 2024.

319 Alexander Matt Turner, Lisa Thiergart, David Udell, Gavin Leech, Ulisse Mini, and Monte Mac-
320 Diarmid. Activation addition: Steering language models without optimization. *arXiv preprint*
321 *arXiv:2308.10248*, 2023.

322 Aäron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive
323 coding. *arXiv preprint arXiv:1807.03748*, 2018.

324 Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov,
325 and Christopher D. Manning. Hotpotqa: A dataset for diverse, explainable multi-hop question
326 answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language*
327 *Processing*, pages 2369–2380, 2018. doi: 10.18653/v1/D18-1259.

328 Huiling Zhen, Weizhe Lin, Renxi Liu, Kai Han, Yiming Li, Yuchuan Tian, Hanting Chen, Xiaoguang
329 Li, Xiaosong Li, Chen Chen, Xianzhi Yu, Mingxuan Yuan, Youliang Yan, Peifeng Qin, Jun Wang,
330 Yu Wang, Dacheng Tao, and Yunhe Wang. DLLM agent: See farther, run faster. *arXiv preprint*
331 *arXiv:2602.07451*, 2026. doi: 10.48550/arXiv.2602.07451.

332 Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan,
333 Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J.
334 Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson,
335 J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI
336 transparency. *arXiv preprint arXiv:2310.01405*, 2023.

337 NeurIPS Paper Checklist

338 **1. Claims** Do the main claims made in the abstract and introduction accurately reflect the paper’s
339 contributions and scope?

340 *Answer:* Yes. The abstract and introduction state the dual-certificate framework, the static upper
341 bound (Proposition 1), the dynamic lower bound (Proposition 2), and the experimental results that
342 correspond to them. All claims are matched by theoretical derivations (§4–5) and empirical evidence
343 (§6).

344 **2. Limitations** Does the paper discuss limitations of the work?

345 *Answer:* Yes. Limitations are discussed in several places: the observational bottleneck and proxy-
346 estimator bias/variance (Appendix C), the scope of claims (§7), the conservative nature of the static
347 bound, and the dependence on software-orthogonality assumptions for tightness (§4, Appendix A).

348 **3. Theory Assumptions and Proofs** Did you state the full set of assumptions of all theoretical
349 results, and did you include complete proofs of all theoretical results?

350 *Answer:* Yes. All assumptions are stated with each proposition (software orthogonality in Lemma 1,
351 conditional Markov and DPI for Proposition 2). Complete proofs are given in the main text (proof
352 sketches) and in full detail in Appendix A. A Lean 4 artifact machine-checks the structural reductions
353 (Appendix D).

354 **4. Experimental Result Reproducibility** Did you provide a way to reproduce the experimental
355 results?

356 *Answer:* Yes. Detailed experimental protocols are described in Appendix B, including the static-
357 certificate computation procedure and the dynamic-certificate estimation steps (proxy, replay, inter-
358 vention, and private-report intervention). The paper references implementation README files. Code,
359 data, and the Lean artifact are included in the supplementary material.

360 **5. Open access to data and code** Did you include the code, data, and instructions needed to
361 reproduce the main experimental results?

362 *Answer:* Yes. Code, precomputed results, and reproduction instructions are provided in the supple-
363 mentary material. The experiments use the open-weight Qwen2.5-7B-Instruct model (Apache 2.0).

364 **6. Experimental Setting/Details** Did you specify all the training details (e.g., data splits, hyperpa-
365 rameters, how they were chosen)?

366 *Answer:* Yes. Experimental details are reported in §6 and Appendix B: e.g., 5-fold cross-fitted logistic
367 regression, trajectory-block bootstrap with 95% CIs, PCA dimensions, audit discretization $Q = 256$,
368 perturbation levels, controller sample count k , and task-pool sampling seed. Hyperparameter choices
369 are motivated and described.

370 **7. Experiment Statistical Significance** Does the paper report error bars suitably and correctly
371 defined or other appropriate information about the statistical significance of the experiments?

372 *Answer:* Yes. All experimental tables report 95% confidence intervals obtained via trajectory- or
373 task-block bootstrap (§6.2–6.4, Appendix C). Error bars are shown in the appendix proxy figure
374 (Figure 3). The method for calculating CIs is explained in the text and Appendix B.

375 **8. Experiments Compute Resource** For each experiment, does the paper provide sufficient
376 information on the computer resources (type of compute workers, memory, time of execution) needed
377 to reproduce the experiments?

378 *Answer:* Yes. The paper states that all pipelines ran on a single Apple M4 Max workstation (§6). This
379 indicates the hardware class and scale; no GPU-cluster requirements are needed.

380 **9. Code Of Ethics** Have you read the NeurIPS Code of Ethics and ensured that your research
381 conforms to it?

382 *Answer:* Yes. The authors have read and adhere to the NeurIPS Code of Ethics. The work focuses on
383 safety-oriented auditing of AI agents and does not involve human subjects, privacy-sensitive data, or
384 dual-use risks beyond those discussed in the paper.

385 **10. Broader Impacts** If appropriate for the scope and focus of your paper, did you discuss potential
386 negative societal impacts of your work?

387 *Answer:* Yes. The paper develops a framework for auditing AI agents, which is inherently aimed
388 at improving transparency and safety. Potential dual-use concerns (e.g., adversaries studying the
389 certificates to design harder-to-audit systems) are indirectly acknowledged in the discussion of
390 limitations and the conservative nature of the bounds (§7). The overall contribution promotes
391 responsible AI deployment.

392 **11. Safeguards** Do you have safeguards in place for responsible release of models with a high risk
393 for misuse (e.g., pretrained language models)?

394 *Answer:* N/A. The paper does not release any new pretrained models. It uses existing open-weight
395 models (Qwen2.5) as-is for auditing experiments without modifying or redistributing them.

396 **12. Licenses** Did you cite the creators and respect the license and terms of use of existing assets?

397 *Answer:* Yes. The paper cites the model and dataset assets where applicable, acknowledges the
398 supporting software communities (Qwen2.5, LLaDA, Hugging Face Transformers, PyTorch, NumPy,
399 scikit-learn, NetworkX, Matplotlib, Lean 4, Lake, and Mathlib; §* Acknowledgments), and uses
400 publicly available assets in accordance with their respective licenses.

401 **13. New Assets** Are you releasing new assets (datasets, code, models), and did you document them
402 properly?

403 *Answer:* No. The experimental code, data, and Lean 4 formalization are provided in the supplementary
404 material for review and will be publicly released and documented upon acceptance.

405 **14. Crowdsourcing and Research with Human Subjects** Did you use crowdsourcing or conduct
406 research with human subjects?

407 *Answer:* N/A. The research does not involve crowdsourcing or human subjects.

408 **15. IRB Approvals** Did you describe potential participant risks and obtain Institutional Review
409 Board (IRB) approvals (or an equivalent approval/review)?

410 *Answer:* N/A. No human subjects research was conducted; therefore, IRB approval is not applicable.

411 **16. Declaration of LLM usage** Does the paper describe the usage of LLMs if it is an important,
412 original, or non-standard component of the core methods in this research?

413 *Answer:* N/A. The LLM (Qwen2.5) serves only as the audit target—it is not a component of the
414 proposed auditing method. The auditing framework itself (min-cut computation, mutual-information
415 lower bounds, and the dual-certificate logic) is independent of any particular language model and
416 does not rely on LLMs as a core methodological component.

A Network Information Theory Background for Proposition 1

Conditional channel capacity. For each unlogged edge e in the time-unrolled DAG G_t , the conditional capacity c_e is the maximum information that can flow through e under the worst-case distribution consistent with the system protocol. It is not an empirical average mutual information. It is a structural upper bound read from interface specifications: token window length, hidden-dimension size, quantization level, and discrete-state cardinality. Concretely, a text channel with budget K tokens over vocabulary $\mathcal{V}$ has $c_e \leq K \log |\mathcal{V}|$; a vector state of dimension d at quantization Q has $c_e \leq d \log Q$; a discrete scheduler with $|\Omega|$ states has $c_e \leq \log |\Omega|$. The $K \log |\mathcal{V}|$ form is the maximum-entropy bound (uniform over K -token sequences); any other distribution gives strictly lower entropy. Auditors who can measure the empirical input distribution to a particular channel may substitute a tighter bound, which only improves the certificate.

What tightening means. The static certificate remains valid under any replacement $c_e^* \geq I(X_e; Y_e | \tilde{T}_t)$, so tightening is a matter of choosing smaller but still defensible per-edge budgets. In practice, this means replacing the worst-case $K \log |\mathcal{V}|$ or $d \log Q$ budget with an empirical or conditional estimate at the same audit discretization, rather than changing the theorem itself.

Cut-set upper bound and additive decomposition. Let $G_t = (V_t, E_t)$ be the time-unrolled DAG. For a cut $\Omega \subseteq V_t$ separating source nodes from a target node, the *induced cut capacity* $C_{\text{cut}}(\Omega)$ is the supremum, over joint input distributions $p(X_\Omega | \tilde{T}_t)$, of the directed conditional mutual information $I(X_\Omega \rightarrow Y_{\Omega^c} | \tilde{T}_t, X_{\Omega^c})$. The network-information-theoretic cut-set upper bound [El Gamal and Kim, 2011] states that any information flowing from sources in Ω to a target in Ω^c is bounded above by $C_{\text{cut}}(\Omega)$, and minimizing over cuts gives the tightest such bound. Proposition 1 applies this bound to the unrecorded trace fragment M_t . We use the cut-set *upper bound* direction only; the network-coding achievability results [Ahlsweide et al., 2000], which concern lower bounds on capacity under cooperative coding across the cut, are not invoked. The induced capacity $C_{\text{cut}}(\Omega)$ is in general a supremum over joint input distributions and cannot be read directly from interface specifications. Lemma 1 shows that under software orthogonality, when the channel laws across the cut factor into independent single-edge channels conditional on $\tilde{T}_t$, the induced capacity is bounded above by the sum of per-edge capacities:

$$C_{\text{cut}}(\Omega) \leq \sum_{e \in E(\Omega, \Omega^c)} c_e.$$

Corollary 1 combines this with Proposition 1 to reduce the certificate computation to a discrete minimum-cut problem on G_t with edge capacities c_e . This is the form used in the static certificate experiment and in the protocol in §B.

Formal derivation of Proposition 1. The proof in §4 has two steps: a trace-gap identity and a cut-set bound via d -separation. We restate the argument here with the d -separation step explicit.

Step 1: Trace-gap identity. Let $M_t := T_t \setminus \tilde{T}_t$. The chain rule of conditional entropy gives

$$H(S_t | \tilde{T}_t) = H(S_t | T_t) + I(S_t; M_t | \tilde{T}_t).$$

The first term is $\varepsilon_{\text{nominal}} = H(S_t | T_t)$.

Step 2: Cut-set bound via d -separation. Fix a cut $\Omega \in \text{Cuts}(U_t \rightarrow S_t)$. By the definition of a cut in the time-unrolled DAG G_t , every directed path from a node in Ω to $S_t \in \Omega^c$ must cross at least one edge in $E(\Omega, \Omega^c)$. The causal influence of M_t on S_t must be mediated by the cross-cut message stream $(X_\Omega \rightarrow Y_{\Omega^c})$ together with the boundary state X_{Ω^c} . Formally, M_t is d -separated from S_t by $(X_\Omega \rightarrow Y_{\Omega^c}, X_{\Omega^c})$ conditional on $\tilde{T}_t$, which yields

$$I(S_t; M_t | \tilde{T}_t) \leq I(X_\Omega \rightarrow Y_{\Omega^c} | \tilde{T}_t, X_{\Omega^c}).$$

By the definition of $C_{\text{cut}}(\Omega)$ as the supremum of the right-hand side over admissible joint input distributions on X_Ω ,

$$I(S_t; M_t | \tilde{T}_t) \leq C_{\text{cut}}(\Omega).$$

459 Since this holds for every cut, it holds for the minimum,

$$I(S_t; M_t \mid \tilde{T}_t) \leq \min_{\Omega \in \text{Cuts}(U_t \rightarrow S_t)} C_{\text{cut}}(\Omega).$$

460 Combining with Step 1 completes the proof of Proposition 1. Corollary 1 follows immediately by sub-
 461 stituting Lemma 1 into each software-orthogonal cut: under orthogonality, $C_{\text{cut}}(\Omega) \leq \sum_{e \in E(\Omega, \Omega^c)} c_e$,
 462 and the minimum over cuts becomes a discrete min-cut problem on the unlogged subgraph with edge
 463 capacities c_e .

464 B Experimental Protocols

465 B.1 Static Certificate Computation Protocol

466 Inputs: deployment architecture specification, logging manifest, and protocol budgets (token window
 467 sizes, hidden dimensions, quantization levels). Procedure: (1) construct the time-unrolled DAG G_t ;
 468 (2) partition edges into logged and unlogged; (3) verify software orthogonality (Lemma 1) or add
 469 shared-state edges explicitly to G_t until orthogonality holds; (4) assign capacity c_e per unlogged
 470 edge using the forms in §4; (5) enumerate reachable cut sets $\mathcal{C}_{\text{unlogged}}$ with a min-cut algorithm after
 471 logged edges are collapsed to zero residual capacity; (6) compute $\varepsilon_{\text{state}}^{\text{UB}} = \varepsilon_{\text{nominal}} + \min_C \sum_{e \in C} c_e$
 472 via Corollary 1.

473 For the logging ablation, repeat steps (2)–(6) for the ordered manifests used in §6.1: output-
 474 only trace, tool-call trace, router-logit trace, summary-buffer trace, and full scratchpad
 475 trace. Report the upper-bound value and active minimum-cut edges at each level. See
 476 experiments/7.1_static_certificate/README.md for implementation.

477 B.2 Dynamic Certificate Estimation Protocol

478 **Proxy certificate.** Inputs: model, task environment, and probe family $Z_t^{(k)} = f_k(S_t)$. Steps:
 479 (1) run inference on N trajectories and record $(\tilde{T}_t, Z_t^{(k)}, A_t)$; (2) train cross-fitted predictors $p_\theta(A_t \mid$
 480 $\tilde{T}_t)$ and $p_\phi(A_t \mid \tilde{T}_t, Z_t^{(k)})$; (3) compute the cross-entropy difference $\hat{I}_k = L_{\text{trace}} - L_{\text{trace}+Z^{(k)}}$;
 481 (4) estimate trajectory-block bootstrap intervals; (5) repeat for random, label-permuted, 1-dimensional,
 482 3-dimensional, 5-dimensional, and full-slice proxies.

483 **Optional controlled replay certificate.** Inputs: model and two controlled execution regimes (wild
 484 and replay). Steps: (1) for each $\tilde{T}_t$, run the same system under wild and replayed missing-state
 485 conditions; (2) estimate $\hat{\Delta} = \hat{\text{JS}}(P_{\text{wild}}, P_{\text{replay}} \mid \tilde{T}_t)$; (3) aggregate over trace slices. This is not
 486 output-only paraphrase replay; it is valid only under the controlled replay condition in §5.

487 **Intervention certificate.** Inputs: model, targeted hidden module, perturbation budget, and task split.
 488 Steps: (1) inject ξ_{hidden} at levels $\{0, \sigma_1, \sigma_2, \dots\}$ while holding $\tilde{T}_t$ fixed; (2) record $(\xi_{\text{hidden}}, A_t \mid \tilde{T}_t)$
 489 pairs; (3) estimate $\hat{I}(\xi_{\text{hidden}}; A_t \mid \tilde{T}_t)$ via conditional action-distribution shift for discrete action
 490 spaces, or via InfoNCE/MINE for continuous action spaces; (4) repeat on dormant tasks and active
 491 tasks under the same topology to test whether $\delta_{\text{act}}^{\text{LB}}$ changes while $\varepsilon_{\text{state}}^{\text{UB}}$ remains fixed.

492 **Multi-agent private-report certificate.** Inputs: task pool, worker model, controller model, and a
 493 two-class API space. Steps: (1) sample $N/2$ arithmetic tasks and $N/2$ factual-retrieval tasks; (2) have
 494 the worker generate a private evidence report under a prompt that forbids direct API names and audit
 495 each report for leakage; (3) sample the controller action distribution k times under wild, neutral
 496 replay, opposite-class report, same-class shuffled report, and oracle-tag conditions; (4) parse actions
 497 into $\{\text{calculator}, \text{search}, \text{invalid}\}$; (5) compute per-task JS divergence in bits and task-block
 498 bootstrap intervals. The oracle-tag contrast is reported only as an upper bound; the main contrast is
 499 the opposite-class worker-report intervention.

500 See the README files for experiments 7.2, 7.3, and 7.6 for implementation.

C Proxy Estimator: Bias and Variance

The read-only proxy experiment is a calibration of observational access, not a main activation result. On 3,000 tool-selection trajectories (mixed calculator/search/email/calendar/weather), we record $\tilde{T}_t$ (embedding-layer representation), Z_t (layer-24 tool-logit projection), and A_t . The CE-diff $\hat{I} = \mathcal{L}(A_t | \tilde{T}_t) - \mathcal{L}(A_t | \tilde{T}_t, Z_t)$ is estimated via 5-fold cross-fitted logistic regression with trajectory-block bootstrap. Table 5 reports the resolution ablation: random, 1/3/5-dim PCA, and full 10-dim proxy.

The $d=5$ PCA proxy recovers 0.0317 bits (1.4% of the $\log_2 5 = 2.32$ bit ceiling) with CI excluding zero on the mixed-task distribution. The magnitude reflects an *observational bottleneck*: when the baseline predictor achieves near-perfect accuracy, the ceiling $H(A_t | \tilde{T}_t) \approx 0$ caps any observational proxy. This is consistent with Pearl’s Causal Hierarchy [Pearl, 2009]: proxy certificates operate at Level 1 (association), while the intervention and replay certificates in §6.2 operate at Level 2 (intervention), breaking the deterministic data-generating process.

The dormant/active proxy split should be read as an estimator-noise diagnostic. On calculator tasks (dormant scratchpad), $H(A_t | \tilde{T}_t) = 0$ empirically, yielding $\delta_{\text{act}}^{\text{LB}} = 0$ trivially. On planning tasks (active scratchpad), the proxy row is positive in point estimate but has a wide CI that straddles zero. We therefore do not use the read-only proxy as the main body evidence for active scratchpad use; the causal intervention and controlled replay in §6.2 supply that evidence.

Proxy	CE-diff (bits; nats)	95% CI (bits)
<i>Resolution ablation (mixed task)</i>		
random	−0.0007; −0.0005	[−0.0012, −0.0003]
$d = 1$ PCA	+0.0065; +0.0045	[+0.0053, +0.0078]
$d = 3$ PCA	+0.0250; +0.0173	[+0.0225, +0.0273]
$d = 5$ PCA	+0.0317; +0.0220	[+0.0273, +0.0358]
full (10-dim)	+0.0183; +0.0127	[+0.0134, +0.0229]
<i>Dormant/active split ($d=64$ PCA, L2-regularized)</i>		
calculator (dormant)	0.0000; 0.0000	[0, 0]
planning (active)	+0.0412; +0.0286	[−0.0725, +0.1386]

Table 5: Read-only proxy calibration. Top: resolution ablation on the mixed tool-selection dataset. Bottom: dormant/active split. The active-task row is reported honestly as an estimator-noise diagnostic because its confidence interval straddles zero; causal replay/intervention supplies the body evidence for active scratchpad use. Negative random-control CE-diff is a finite-sample artifact near the noise floor and is clipped to zero when interpreted as a lower-bound certificate.

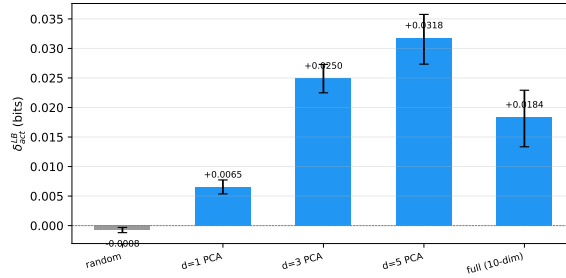


Figure 3: Proxy resolution ablation. $\delta_{\text{act}}^{\text{LB}}$ increases monotonically through $d=5$ PCA. The random proxy sits near zero (estimator leakage check). Error bars are 95% trajectory-block bootstrap CIs.

The CE-diff estimator $\hat{I} = L(\tilde{T}_t) - L(\tilde{T}_t, Z_t)$ lower-bounds $I(Z_t; A_t | \tilde{T}_t)$, but three bias sources and two variance sources matter in practice.

Bias sources. (1) *Proxy coarsening*: $Z_t = f(S_t)$ discards information in S_t not captured by f , so $I(Z_t; A_t | \tilde{T}_t) \leq I(S_t; A_t | \tilde{T}_t)$. The gap is the coarsening loss. (2) *Restricted predictor class*:

Logistic regression or shallow MLPs may not reach the Bayes-optimal predictor, so the CE-diff underestimates the true conditional MI. (3) *Finite-sample bias*: With finite N , the cross-entropy on held-out folds is an upward-biased estimate of the population CE, and the difference of two such estimates carries residual bias. Cross-fitting mitigates but does not eliminate this.

Variance sources. (1) *Trajectory sampling*: The N observed trajectories are a finite sample from the deployment distribution. Block bootstrap over trajectories estimates the resulting CI. (2) *Proxy dimension*: Higher-dimensional proxies carry more information but increase predictor variance. The PCA ablation in Table 5 shows this trade-off: $d=5$ outperforms both $d=3$ and the full 10-dimensional proxy.

Mean squared error. Decomposing $\text{MSE} = (\text{bias})^2 + \text{variance}$, the proxy certificate is a conservative lower bound: positive bias (underestimation) is safe for the audit (it never overstates δ_{act}), while variance widens the CI but does not shift the point estimate.

Sampling noise and temperature. Increasing decoding temperature can raise $H(A_t \mid \tilde{T}_t)$, but independent sampler noise is not a certificate amplifier. It changes the audited policy and can reduce $I(Z_t; A_t \mid \tilde{T}_t)$ when the added action variation is unrelated to the hidden state proxy. A temperature sweep is therefore a calibration of a randomized deployment policy, not a substitute for proxy quality or causal access. We do not use temperature injection for the main proxy certificate; the synthetic experiment instead injects hidden-state stochasticity with known ground truth.

D Lean 4 Formalization Boundary

The Lean 4 artifact mechanizes the certificate reductions, the end-to-end case study, and the structural impossibility and identifiability results. The main theorem files are `DualCertificate.lean` (static and dynamic certificates), `CutSetBoundExtract.lean` (cut-set DPI bottleneck reduction), `ChannelCapacity.lean` (KKT certificate framework), `CaseStudy.lean` (linear chain end-to-end bound), `Screenability.lean` (autoregressive zero-cut impossibility), and `InternalImpossibility.lean` (predictability-route impossibility). `IdentifiabilityGap.lean` gives an axiom-free construction of behaviorally equivalent but audit-inequivalent finite PMFs (Theorem 1). Full details are given in the verification `README.md`.

The artifact uses an axiomatic interface: the structural reductions are machine-checked, while the cut-set capacity inequality is imported as an axiom and consumed by the downstream propositions marked C/E in Table 6 (see §D.1 for the precise boundary). Every result marked C—entropy and mutual-information non-negativity, the trace-gap chain rule, the conditional Markov property, the conditional DPI, the autoregressive zero-cut impossibility, the predictability-route impossibility, and the identifiability-gap construction—is proved from Mathlib first principles using finite-discrete entropy formulas over `FinitePMF`.

D.1 External / Assumed Steps

The following is imported as an explicit Lean axiom and consumed by the downstream machine-checked propositions marked C/E in Table 6:

- **Cut-set capacity inequality**: the structural premise that $I(S; M \mid \tilde{T}_t) \leq C_{\text{cut}}(\Omega)$ for the chosen cut Ω , consumed by `prop1_static_ub`, `abstract_cut_set_bound`, and all C/E items in Table 6. In the linear chain case study this bound is discharged by the KKT certificate pipeline via `capacity_le_of_kkt`.
- **Conditional Markovity**: the premise `condMarkov` in `prop2_dynamic_lb`, `cut_set_dpi_bound`, and the case study. This is a concrete algebraic predicate on finite PMFs, not an axiom; the DAG automation stages (Stages 1–2 in the verification `README`) are designed to derive it from the deployment topology.

The entropy and mutual information non-negativity, the trace-gap chain rule, the conditional Markov property (`condMarkov`), the finite conditional Data Processing Inequality (`cond_dpi`), the autoregressive zero-cut impossibility, the predictability-route impossibility, and the identifiability-gap

Main-Text Item	Lean File	Lean Theorem	Status
Entropy, KL, CMI non-negativity	InfoTheory	entropy_nonneg,	C
Trace-gap chain rule	InfoTheory / InfoTheoryHelpers	condMutualInfo_nonneg static_decomposition, chain_rule_I_A_BC	C
Conditional DPI (finite)	InfoTheory	cond_dpi	C
Conditional Markov property	InfoTheory	condMarkov	C
Cor 2 (Autoregressive zero-cut)	Screenability	no_eis_autoregressive	C
Cor 2 (Predictability route)	InternalImpossibility	internal_impossibility_predictability	C
Lemma 1 (Additive decomposition)	DualCertificate	static_decomposition, software_orthogonal	C
Static cardinality corollary	DualCertificate	prop1_static_ub_bounded	C
Prop 1 (General cut-sum bound)	DualCertificate	prop1_static_ub	C/E
Cor 1 (Orthogonal min-cut)	DualCertificate	corollary_additive_ub	C/E
Cut-set DPI bottleneck	CutSetBoundExtract	abstract_cut_set_bound	C/E
KKT certificate bound	ChannelCapacity	capacity_le_of_kkt	C/E
Linear chain bound	CaseStudy	linear_chain_cut_set_bound	C/E
($\varepsilon_{\text{state}}^{\text{UB}} \leq 1$ bit)			
Prop 2 (Conditional DPI certificate)	DualCertificate	prop2_dynamic_lb, aggregated_dynamic_lb	C/E
Thm 1 (Identifiability gap)	IdentifiabilityGap	identifiability_gap_extremes	C

Table 6: C = machine-checked from Mathlib first principles with no external premises. C/E = structural reduction is machine-checked conditional on the cut-set capacity axiom (§D.1). Entropy, conditional entropy, and conditional mutual information are defined by finite-discrete formulas using Mathlib finite sums and real logarithms. The trace-gap chain rule, conditional Markov property, and conditional DPI are proved from these definitions; only the cut-set capacity inequality remains external.

571 construction (all marked C in Table 6) are *not* axioms: they are proved from Mathlib first principles
572 using finite-discrete entropy formulas over `FinitePMF`. The artifact verifies that the main-text propo-
573 sitions follow from the cut-set premise by elementary algebraic manipulation. It does not derive the
574 cut-set bound itself from Lean’s measure-theoretic foundations.

575 E Synthetic Ground-Truth Validation

576 To test whether the certificates recover known hidden influence, we construct a synthetic agent whose
577 hidden-state entropy and decision relevance are controlled by design. A binary hidden variable
578 $H \sim \text{Bern}(0.5)$ injects influence of strength β_h into the action logits; the unlogged-channel capacity
579 $c_H = H(H) = 1$ bit is known. We run both the static min-cut ($\varepsilon_{\text{state}}^{\text{UB}} = 1$ bit) and the proxy CE-diff
580 estimator on 1,000 trajectories at five influence levels.

581 Fig. 4 shows $\varepsilon_{\text{state}}^{\text{UB}} = 1$ bit matches the true hidden entropy (the bound is tight when there is no
582 redundant unlogged capacity). The dynamic lower bound $\delta_{\text{act}}^{\text{LB}}$ tracks the true conditional MI from
583 below at all tested influence levels ($\beta_h \geq 1$); negative CE-diff values at $\beta_h = 0$ reflect finite-sample
584 variance around a near-zero true MI.

585 The Qwen deployment has a near-deterministic baseline that suppresses the proxy signal ($H(A_t |$
586 $\tilde{T}_t) \approx 0$). The synthetic environment instead injects controlled stochasticity ($H = 1$ bit of hidden-
587 state entropy). Without that observational bottleneck, the proxy certificate $\delta_{\text{act}}^{\text{LB}}$ tracks the true
588 conditional-MI curve from below in the tested range. The validation shows that the proxy certificate
589 can become tight when structural capacity is behaviorally active *and* the observational ceiling permits
590 detection.

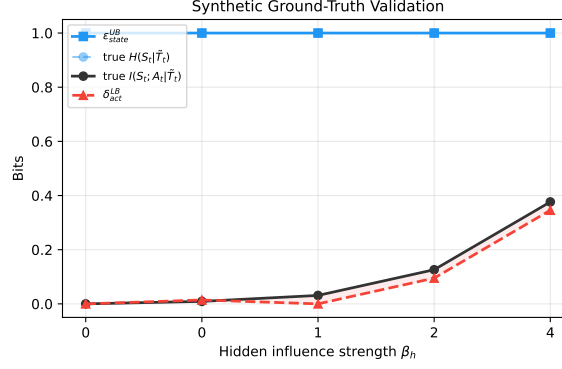


Figure 4: Synthetic ground-truth. $\epsilon_{\text{state}}^{\text{UB}}$ (blue) equals the true $H(S_t | \tilde{T}_t) = 1$ bit. $\delta_{\text{act}}^{\text{LB}}$ (red) tracks the true $I(S_t; A_t | \tilde{T}_t)$ from below, with negative values at low β_h reflecting estimator variance near the noise floor.

591 F Tightening the Static Certificate

592 The reported static certificate is intentionally conservative. Its role is to upper-bound the residual
 593 hidden-state entropy from deployment topology alone, so the default budgets in §4 are worst-case
 594 interface capacities rather than empirical information estimates.

595 **Empirical and conditional budgets.** A tighter deployment-specific certificate can replace the
 596 worst-case budget c_e on each unlogged edge by an empirically justified budget c_e^* . For a text channel,
 597 one option is a high-confidence upper bound on $H(X_e | \tilde{T}_t)$; when the visible history strongly
 598 constrains the missing input, a conditional form such as $H(X_e | \tilde{T}_t, \mathcal{H}_t)$ can be smaller still. For
 599 quantized vector states, the analogous replacement is an empirical entropy or rate-distortion budget at
 600 the same audit discretization. These substitutions do not alter the min-cut proof: they only change the
 601 edge weights used by Corollary 1.

602 **Graph refinement.** If two apparent channels share latent state, naive edge-wise additivity can
 603 double-count the same information. The remedy is to refine the time-unrolled DAG so that the shared
 604 state appears as an explicit node or edge family, then recompute the min-cut on the refined graph.
 605 This usually tightens the certificate more reliably than trying to compensate with ad hoc weight
 606 corrections.

607 **Reporting policy.** For review and deployment, report a ladder of certificates: worst-case structural,
 608 empirical-marginal, and empirical-conditional. The first is the theorem-backed safety baseline.
 609 The latter two are optional tightening modes and should be reported only when the underlying
 610 measurements are defensible.